

Examining Danish Housing Sales to Estimate Property Values

Description / Motivation

In the wake of a regional banking crisis in the 1980s, Denmark enacted regulations requiring centralized, digital record-keeping of housing sales. This dataset is a product of those modernization efforts and captures Danish real estate sales from 1992 onward. It serves as the basis for examining the factors associated with residential sale prices and developing a machine learning model to estimate property values.

Research Questions

- Which property characteristics are most associated with Danish residential sale prices?
- Using machine learning models, can these characteristics be used to predict home prices?

Methodology

Data Preparation

The dataset “Danish Residential Housing Prices 1992-2024” was found on Kaggle. It contains ≈1.5 million historical housing sales and 19 variables about the homes sold, including geographical, financial, and descriptive details. Exploratory analysis revealed that sales from 2024 were incomplete because the dataset was compiled after 2024-Q3. The sales from 2024 were repurposed as an unseen holdout dataset for evaluating model performance.

Data Cleaning

Sales types were limited to “regular sales” after identifying it as the dominant method of home sales (Chart A). Redundant location variables City, Area, and Region were removed in favor of the more specific postal code. Price per Square Meter was removed because it is derived from the target variable (Purchase Price), which would inevitably create target leakage and biased model evaluations.

ML Modeling

Random Forest Regression was initially selected but proved computationally inefficient for this dataset. XGBoost was then evaluated, producing an initial R^2 of 0.468. Deliberate feature selection and hyperparameter tuning ultimately improved model performance to an R^2 of 0.7036 on the unseen 2024 holdout dataset.

Exploratory Data Analysis (EDA)

From the same initial investigation that revealed regular sales as the dominant sales type, it was observed that auctions became more noticeable in the years after the global financial crisis of 2008-2009. The crisis led to increased foreclosures, which were subsequently sold at auction. While this made for an interesting observation, Auctions, Family, and Other Sales did not represent a large enough proportion of transactions to be suitable for modeling.

Villas made up more than half the home sales in the dataset. While they are called villas, they can also be considered single-family homes. Apartments, Townhouses, Farms, and Summer Houses combine to account for nearly half the dataset, with each possessing sufficient observations to be valuable for modeling (Chart B).

Because the dataset included a variable defining quarters of sales, a natural question of seasonality arose. An initial review, represented by Chart C, shows that Q2 was busier overall than other quarters. A closer look (Chart D) shows that the second quarter has a buffer between it and the other quarters, but they all follow the same general path from 1992 through 2023. Likewise, average sales prices stay in line year-by-year across the quarters (Chart E). This corresponds with the steady quarterly relationship of sales counts seen in Chart D. The differences between quarters remain relatively minimal and do not meaningfully impact pricing or sales volume, rendering the question of seasonality moot.

A significant observation from exploratory analysis was that price per square meter showed a peak in prices buyers would pay per square meter. Chart F shows a dense triangular cluster of sales points. This leading edge indicates a steady maximum price/m² until the sales points begin to scatter. The resulting gaps indicate a threshold for diminishing returns, where buyers become less willing or able to pay more for larger living spaces. Chart G shows how this has played out for each house type, with differing points of diminishing returns. The superimposed box-and-whisker plots highlight the distribution of home prices. When viewed alongside the scatterplots, the range of the most common housing sizes can be deduced for each housing type.

Despite Chart F showing a steady representation of homes up to roughly 500 m², Chart I demonstrates the minuscule number of large residences sold between 1992 and 2023. Various upper limits were examined before a maximum size of 350 m² was selected. This represents 99.59% of regular residential sales, maintaining the model's scope while reducing the influence outliers have on model performance.

Analysis

The predictive model's estimates were more narrowly focused than actual sales. Charts J – K show the model clustered estimates more acutely than actual sales in the scatterplots of price vs square meter. This can lead to an inference that the model adheres to a price per square meter algorithm to make predictions. Each prediction chart shows an empty space between the plots and the X-axis, while the actual 2024 sales charts show plots covering the same spaces. This could very well be due to the model's inability to tell if a property is nice, average, or poor quality, as this is subjective. Basic introspection reveals that people will pay more for nicer amenities or more convenience. Likewise, some homes fall into disrepair, leading to lower prices per square meter.

It was observed that the model behaved differently at different price ranges, overestimating sales prices at lower levels, and underestimating them at higher ones (Chart L). An explanation

for this could be the same as discussed in the previous paragraph, namely that with the available data, a model cannot quantify quality. To do so would likely require extensive documentation of what components were used to build or update homes and when, for renovations. Again, Charts J-K show wedges of negative space where no lower-price predictions were made beyond various square meter limits.

A number of price ranges were evaluated to establish the median and mean deviations between actual and predicted purchase prices on the 2024 holdout dataset. These price differences were used to adjust price estimates for an online interactive chart.

Findings

As the cliché goes, real estate is all about “location, location, location,” and this was found to be the case with Danish residential sales between 1992 and 2024. By far, the biggest factor affecting price estimates was the postal code of each home. The highest prices per square meter were found in the largest city, Copenhagen, which also serves as Denmark’s capital.

Housing type also mattered in determining price. Apartments have inherent spatial limits due to sharing buildings with other apartments and/or businesses. Additionally, they are increasingly common in cities, where space comes at a premium. These are likely the reasons Apartments were the house type that showed the highest average price per square meter.

Home price predictions appeared to derive from an algorithm centered on price per square meter. This observation is based on Chart K, where predictions stay close to trend lines, particularly on the lower side. By comparison, actual 2024 sales are spread farther from trend lines for each house type. As discussed earlier, this could be due to the data not containing qualitative data about properties. With more detailed data, it is feasible that the model could differentiate between higher, lower, and middling homes.

Limitations

The dataset was quite robust but had some gaps in data collection. 2017 was missing half the year, and 2024 was missing the final quarter of the year. Neither absence was catastrophic for modeling, but more complete data would have been preferred. Some variables didn’t seem to have been input consistently, with the number of rooms being particularly hard to understand.

A factor that likely affected predictions was the home buying behavior during the Covid-19 pandemic. This resulted in a substantial spike in the number of home sales during 2020 (Chart D), and volatility in average price (Chart E). The data is not far enough removed from this period for new, consistent price trends to appear (Chart D). Once enough time passes, updates to the dataset could lead to more accurate predictions. Likewise, qualitative variables might allow the model to make less restrained housing price estimates.

Conclusion

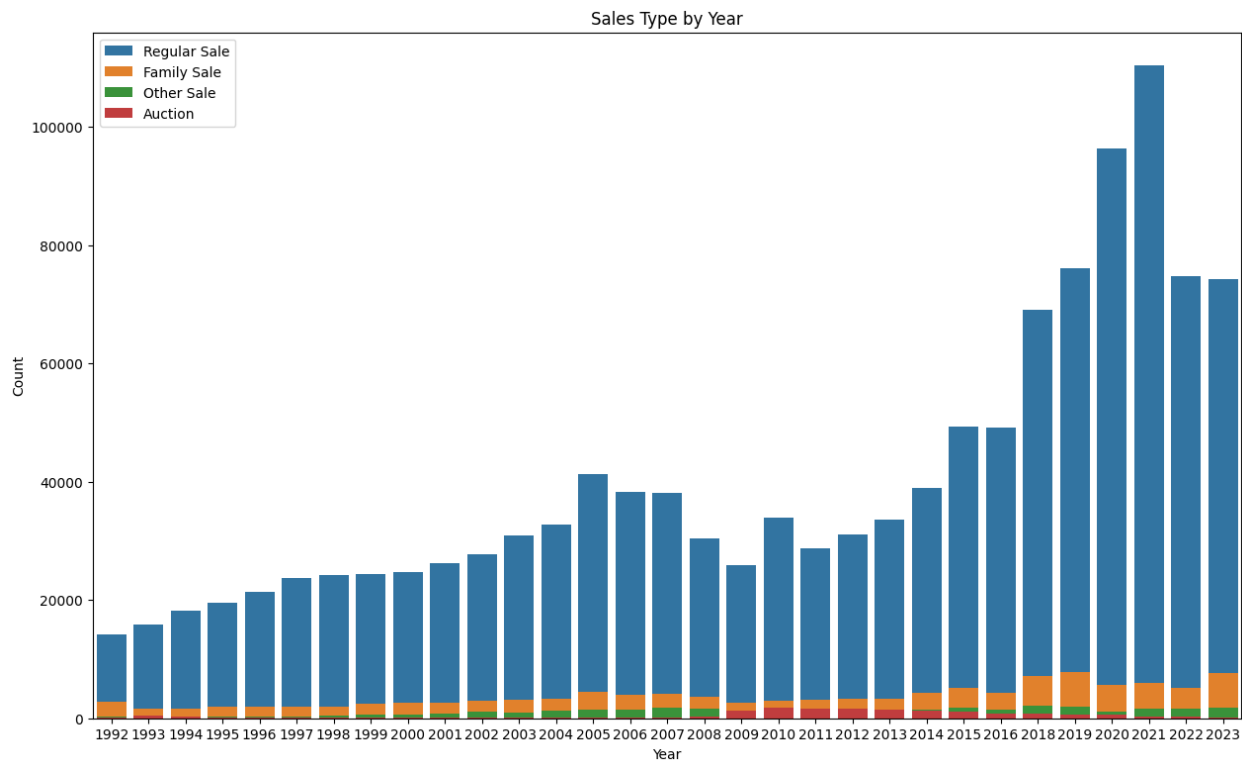
Location proved to hold the most sway over Danish housing values, with postal codes influencing price estimates more than any other variable. House type and size of living area also impacted prices, but the analysis showed that no combination of available details could fully account for the qualitative or subjective factors that determine real-world purchase prices.

Restricting the XGBoost model training to homes of 350 m² or less retained 99.59% of regular residential sales while improving predictive performance, with an R^2 of 0.7036. The model followed the same general price-per-square-meter relationships as actual 2024 sales, but produced a narrower range of estimates.

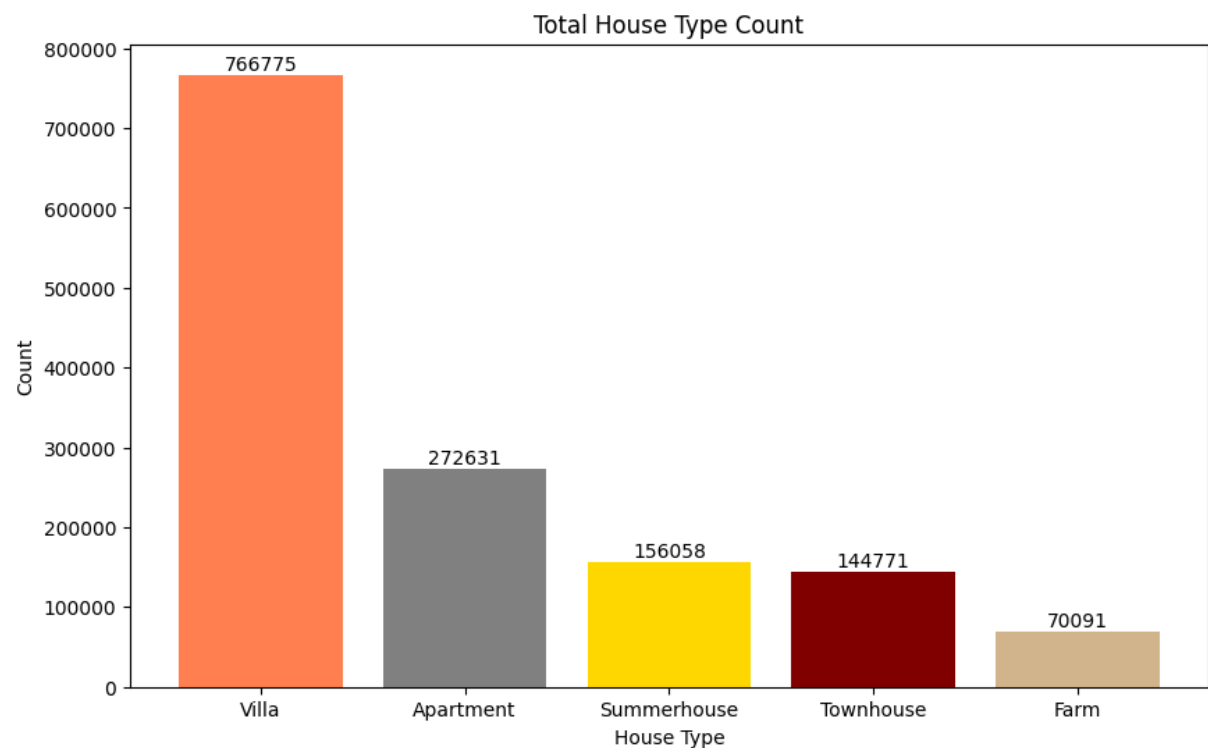
Overall, the project illustrates the predictive ability of machine learning in estimating residential property values from common housing characteristics. While the model could not account for every factor that influences home prices, it establishes reasonable baseline valuations while underlining both the capabilities and limitations of data-driven price estimation.

Charts and Tables

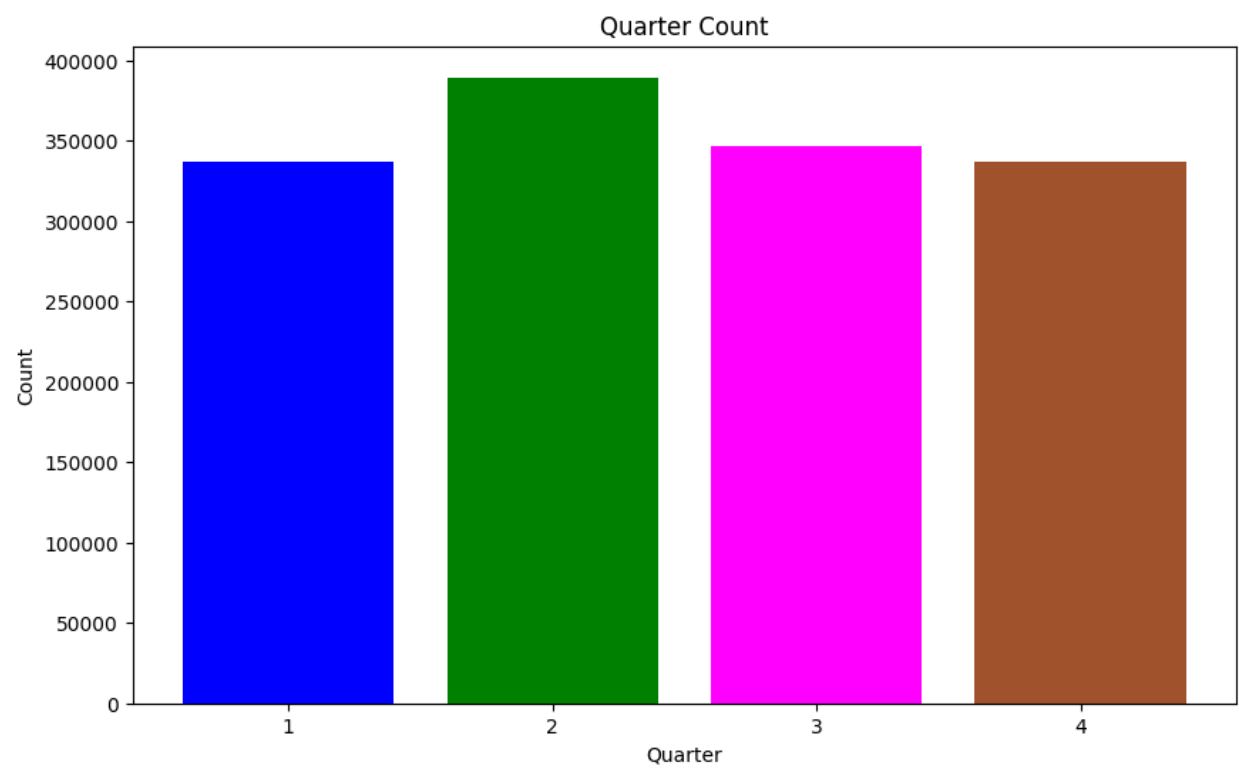
A. Method of Residential Sales



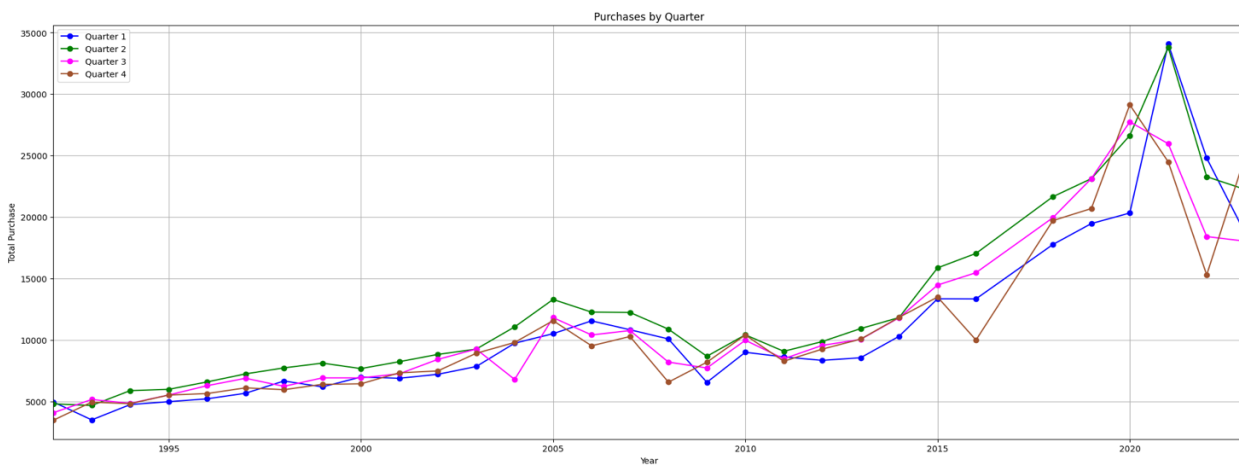
B. Distribution of House Types



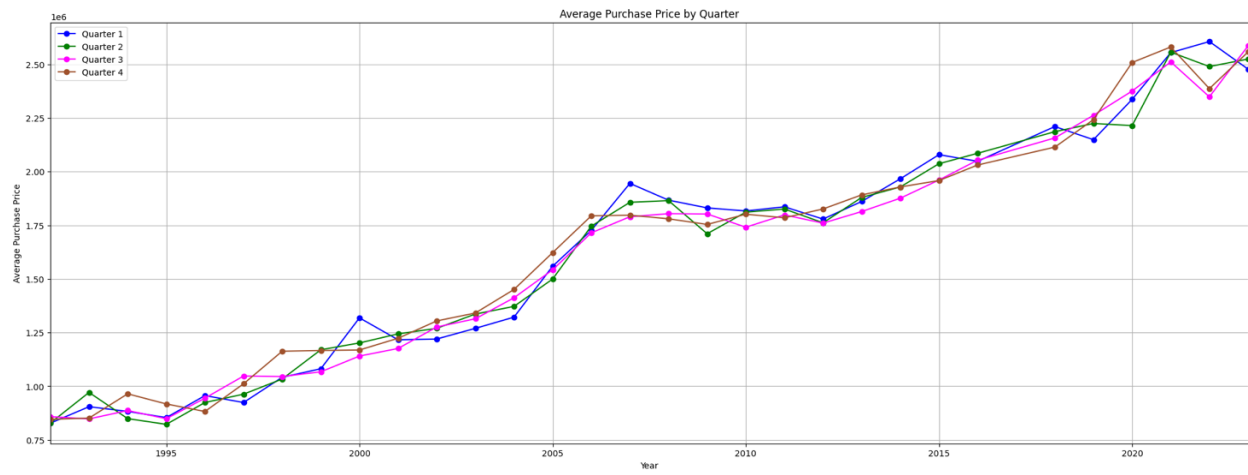
C. Distribution of Quarterly Sales



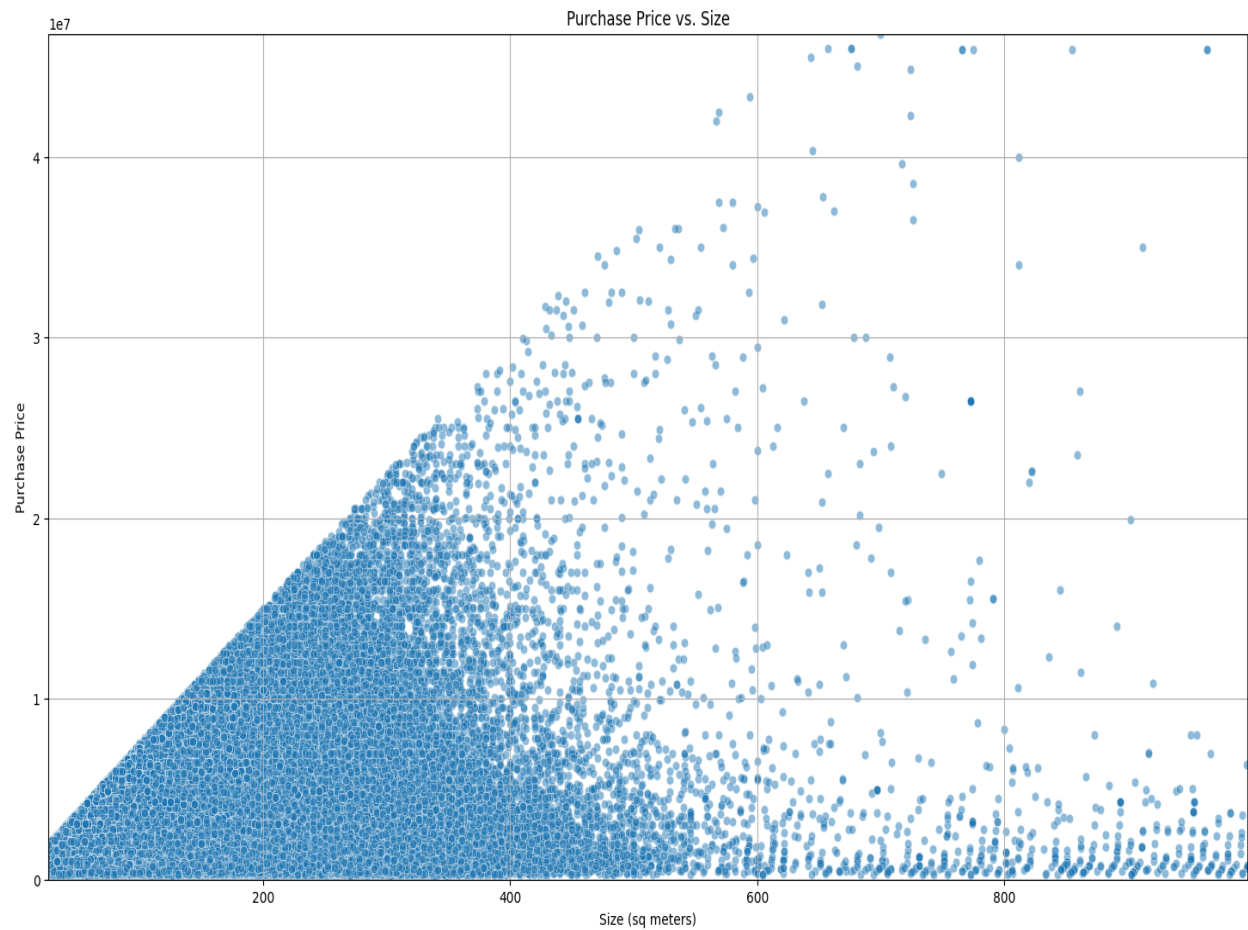
D. Historic Sales Counts



E. Average Quarterly Sales Price 1992 – 2023

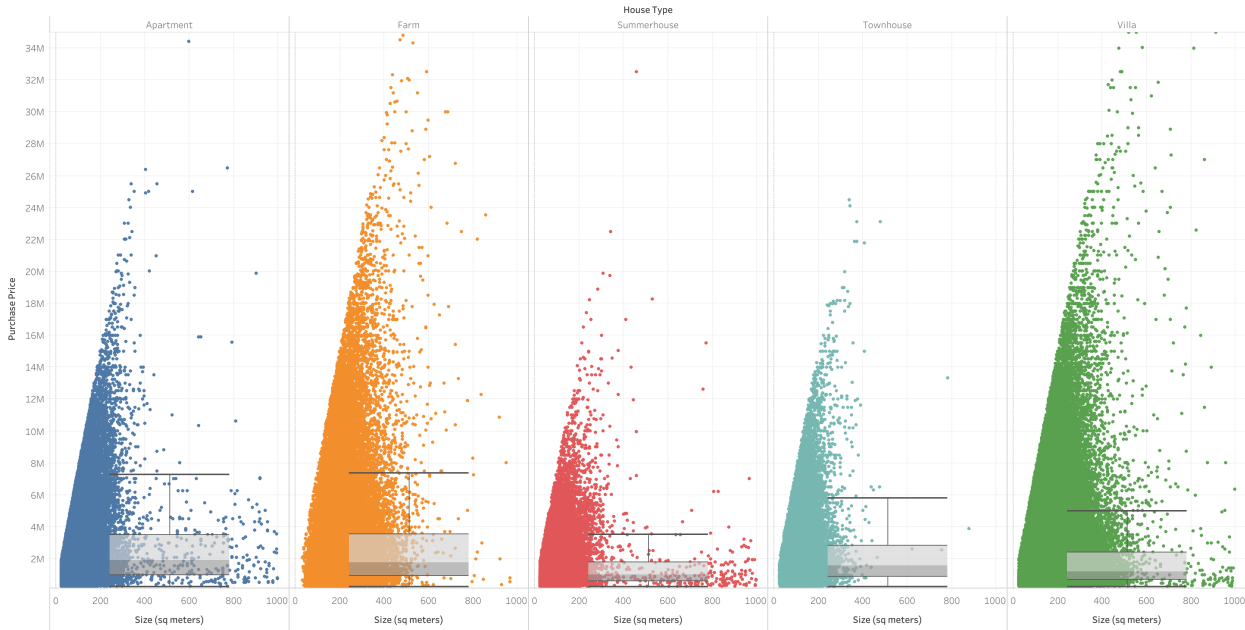


F. Price vs Living Area (Combined)

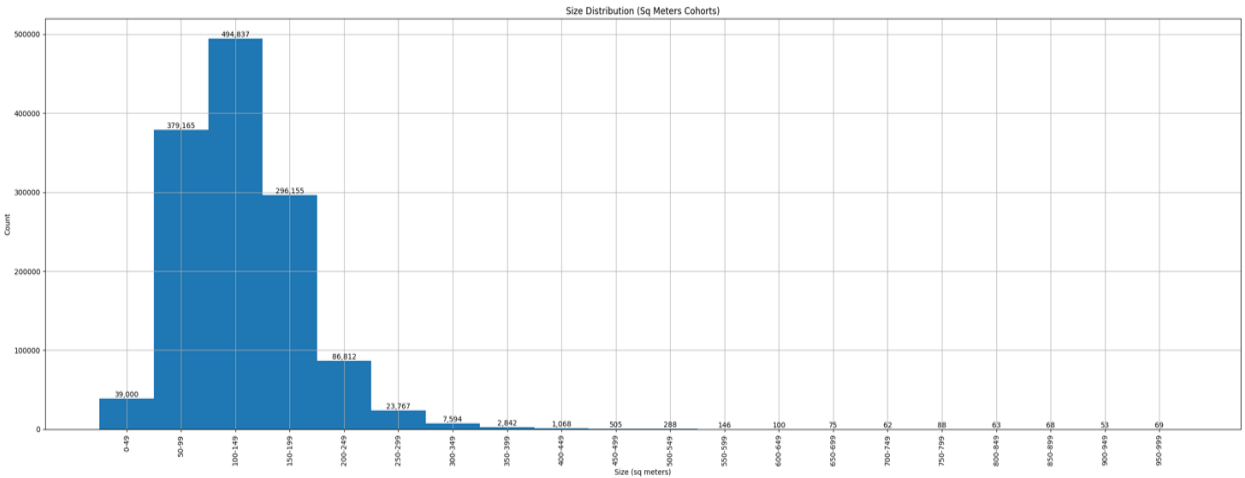


G. Price vs Living Area by House Type

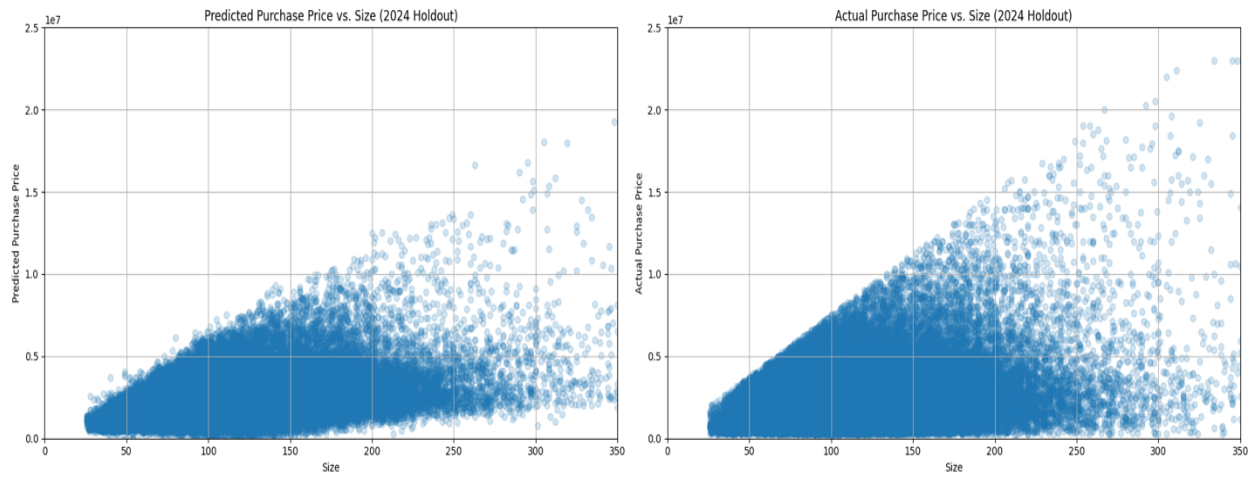
Price vs Living Area across House Types



I. Distribution of Home Sizes



J. Predicted Purchase Prices vs Actual Purchase Prices



K. Actual and predicted purchase prices by living area and house type (2024 holdout)

Actual & Predicted Price vs Square Meter (2024 Holdout)



L. Predicted vs Actual Prices

